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SUBSTRATE, LIGHT-EMITTING DEVICE, METHOD OF MANUFACTURING SUBSTRATE, AND METHOD OF MANUFACTURING LIGHT-EMITTING DEVICE

Abstract

A substrate includes a conductive member including first to third regions and a ceramic member. The first region contains copper and diamond. The second region contains copper as a main component. The third region is disposed between the first and second regions and the ceramic member, and contains copper and at least one selected from the group consisting of titanium, hafnium, zirconium, niobium, cerium, and magnesium. The diamond disposed in the first region in a vicinity of the third region is in contact with the third region. In a cross section, the first region includes a first portion and a second portion located between the first portion and the second region in a first direction, and an area occupied by diamond per unit area in the second portion is smaller than an area occupied by diamond per unit area in the first portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-030351, filed Feb. 29, 2024, the contents of which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a substrate, a light-emitting device, a method of manufacturing a substrate, and a method of manufacturing a light-emitting device.

2. Description of Related Art

[0003] There are substrates including a conductive member containing copper and a ceramic member. It is desirable that the difference in thermal expansion coefficient between the conductive member and the ceramic member of the substrate is small (for example, see Japanese Patent Publication No. 2008-004760.)

SUMMARY

[0004] An embodiment of the present disclosure can advantageously provide a substrate in which a difference between thermal expansion coefficient of a conductive member and that of a ceramic member can be further reduced, and a light-emitting device including the substrate.

[0005] A substrate according to an embodiment includes a conductive member including a first region containing copper and diamond, a second region located on the first region and contains copper as a main component, and a third region; and a ceramic member disposed away from the first region and the second region in a second direction perpendicular to a first direction, the first direction being a direction from the first region toward the second region. The third region is located between the first region and the ceramic member and between the second region and the ceramic member, and contains copper and at least one selected from the group consisting of titanium, hafnium, zirconium, niobium, cerium, and magnesium. The diamond disposed in the first region in the vicinity of the third region is in contact with the third region. In a cross section parallel to the first direction, the first region comprises a first portion and a second portion that is located between the first portion and the second region in the first direction, and an area occupied by the diamond per unit area in the second portion is smaller than an area occupied by the diamond per unit area in the first portion.

[0006] A method of manufacturing a substrate according to an embodiment includes preparing a sintered body having a plurality of protrusions, by disposing a first raw material containing copper powder and diamond inside recesses of a jig having an upper surface and the recesses each recessed from the upper surface, disposing a second raw material containing copper powder on the first raw material and the upper surface of the jig, and sintering the first raw material and the second raw material; disposing, via an active metal brazing material, a ceramic member on a lateral surface of each of the plurality of protrusions of the sintered body removed from the jig; and bonding the ceramic member and the sintered body by sintering the active metal brazing material.

[0007] According to certain embodiments of the present disclosure, it is possible to provide a substrate in which a difference between thermal expansion coefficient of a conductive member and that of a ceramic member can be reduced, and a light-emitting device including the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] A more complete appreciation of embodiments of the invention and many of the attendant advantages thereof will be readily obtained by reference to the following detailed description when considered in connection with the accompanying drawings.

[0009] FIG. 1 is a cross-sectional view illustrating a substrate according to an embodiment.

[0010] FIG. 2 is a plan view illustrating the substrate according to the embodiment.

[0011] FIG. 3 is a bottom view illustrating the substrate according to the embodiment.

[0012] FIG. 4A is an enlarged cross-sectional view of the vicinity of a boundary between a first region and a third region.

[0013] FIG. 4B is a cross-sectional view illustrating an example of a diamond.

[0014] FIG. 4C is a cross-sectional view illustrating an example of a composite particle.

[0015] FIG. 4D is a cross-sectional view illustrating an example of a composite particle.

[0016] FIG. 5 is an enlarged cross-sectional view of the first region and a second region.

[0017] FIG. 6 is an enlarged cross-sectional view of a ceramic member.

[0018] FIG. 7 is a cross-sectional view illustrating a light-emitting device according to the embodiment.

[0019] FIG. 8A is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0020] FIG. 8B is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0021] FIG. 8C is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0022] FIG. 8D is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0023] FIG. 9A is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0024] FIG. 9B is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0025] FIG. 9C is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0026] FIG. 10A is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0027] FIG. 10B is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0028] FIG. 10C is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0029] FIG. 11A is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0030] FIG. 11B is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0031] FIG. 11C is a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0032] FIG. 12A is a schematic view illustrating a step of manufacturing the light-emitting device according to the embodiment.

[0033] FIG. 12B is a schematic view illustrating a step of manufacturing the light-emitting device according to the embodiment.

[0034] FIG. 12C is a schematic view illustrating a step of manufacturing the light-emitting device

according to the embodiment.

[0035] FIG. **12D** is a schematic view illustrating a step of manufacturing the light-emitting device according to the embodiment.

[0036] FIG. **13A** is a cross-sectional view illustrating a substrate according to a variation of the embodiment.

[0037] FIG. **13B** is a cross-sectional view illustrating a substrate according to a variation of the embodiment.

[0038] FIG. **13C** is a cross-sectional view illustrating a substrate according to a variation of the embodiment.

DETAILED DESCRIPTION

Description of Embodiments

[0039] Each embodiment of the present disclosure will be described below with reference to the drawings. The drawings are schematic or conceptual, and the relationships between thicknesses and widths of portions, the ratios of sizes between portions, and the like are not necessarily the same as the actual values thereof. In addition, even in the case of representing the same portion, dimensions or ratios may be represented differently depending on the drawings. In the present specification and the drawings, the same elements as those already described are denoted by the same reference numerals, and detailed description thereof will be omitted as appropriate. As a cross-sectional view, an end view illustrating only a cut surface may be used.

[0040] FIG. **1** is a cross-sectional view illustrating a substrate according to an embodiment. FIG. **2** is a plan view illustrating the substrate according to the embodiment. FIG. **3** is a bottom view illustrating the substrate according to the embodiment. FIG. **1** corresponds to a cross-sectional view taken along line I-I in FIGS. **2** and **3**.

[0041] As illustrated in FIGS. **1** to **3**, a substrate **1** according to the embodiment includes at least one conductive member **10** and a ceramic member **20**. As illustrated in FIG. **1**, each of the at least one conductive member **10** includes a first region **11**, a second region **12**, and a third region **13**.

[0042] In the following description, an XYZ orthogonal coordinate system is used. A direction from the first region **11** toward the second region **12** is referred to as a Z direction (first direction). Two directions perpendicular to the Z direction and orthogonal to each other are referred to as an X direction (second direction) and a Y direction. For the sake of description, the direction from the first region **11** toward the second region **12** is referred to as “upward”, and the opposite direction is referred to as “downward”. These directions are based on the relative positional relationship between the first region **11** and the second region **12**, and are irrelevant to the direction of gravity. The expression “in a plan view” as used in the embodiment refers to viewing the object from above. In the present specification, not only a portion that is directly visible from above, but also a portion that is not directly visible from above may be described as if it is seen through using the expression “in a plan view”.

[0043] The first region **11** contains copper and diamond. The second region **12** is disposed above the first region **11**. The second region **12** contains copper as a main component. The term “main component” refers to a component contained in an amount of 50% or more.

[0044] The ceramic member **20** is disposed away from the first region **11** and the second region **12** in the X direction. The ceramic member **20** has insulating properties. The ceramic member **20** can be, for example, a sintered body of silicon nitride, aluminum nitride, aluminum oxide, zirconium oxide, or the like.

[0045] The third region **13** is disposed between the first region **11** and the ceramic member **20** and between the second region **12** and the ceramic member **20** in the X direction. The third region **13** is in contact with the first region **11**, the second region **12**, and the ceramic member **20**. Therefore, in the example illustrated in FIG. **1**, the conductive member **10** and the ceramic member **20** are in contact with each other. As illustrated in FIG. **1**, the third region **13** is further located below the first region **11**. The third region **13** is further located above a portion of the ceramic member **20**.

[0046] The third region **13** contains copper and at least one selected from the group consisting of titanium, hafnium, zirconium, niobium, cerium, and magnesium. Titanium, hafnium, zirconium, niobium, cerium, and magnesium are so-called active metals. The third region **13** contains copper as a main component and contains an active metal. The third region **13** can further contain silver. When the third region **13** contains silver, the boundary between the first region **11** and the third region **13** is more easily recognized. The third region **13** can further contain tin, indium, or the like.

[0047] As illustrated in FIGS. **1** and **2**, the second region **12**, the third region **13**, and the ceramic member **20** are exposed on the upper surface of the substrate **1**. As illustrated in FIGS. **1** and **3**, the third region **13** and the ceramic member **20** are exposed on the lower surface of the substrate **1**.

[0048] As illustrated in FIG. **1**, the thickness of the first region **11** in the Z direction is greater than the thickness of the second region **12** in the Z direction. With this structure, the proportion of the first region **11** in the conductive member **10** can be higher than the proportion of the second region **12** in the conductive member **10**, so that the thermal expansion coefficient of the conductive member **10** can be further reduced. More preferably, the thickness of the first region **11** in the Z direction is not less than 1.5 times and not more than 50 times the thickness of the second region **12** in the Z direction.

[0049] The dimensional relationship between the first region **11**, the second region **12**, and the ceramic member **20** can be appropriately designed. In one example, the length of the first region **11** in the X direction and the length of the second region **12** in the X direction are greater than the length of the ceramic member **20** in the X direction. This structure allows for improving the heat dissipation of the substrate **1**.

[0050] In the example illustrated in FIGS. **1** to **3**, the substrate **1** includes a plurality of conductive members **10**. The ceramic member **20** is disposed between two conductive members **10** in the X direction. Specifically, one ceramic member **20** having a rectangular shape in a plan view is located between two conductive members **10** having a rectangular shape in a plan view, and the conductive members **10** and the ceramic member **20** are disposed adjacent to each other.

[0051] FIG. **4A** is an enlarged cross-sectional view of the vicinity of a boundary between the first region and the third region. FIG. **4B** is a cross-sectional view illustrating an example of a diamond. FIG. **4C** is a cross-sectional view illustrating an example of a composite particle. FIG. **4D** is a cross-sectional view illustrating an example of a composite particle.

[0052] In the first region **11**, particulate diamond D is dispersed in a base material containing copper as a main component. As illustrated in FIG. **4A**, the diamond D in the first region **11** disposed in the vicinity of the third region **13** is in contact with the third region **13**.

[0053] The diamond D that is used in the present embodiment can be an artificially produced industrial diamond. The diamond D has a higher hardness and a lower thermal expansion coefficient than those of a metal used for the substrate. Therefore, with the diamond D disposed in the conductive member **10**, it is possible to reduce the occurrence of shrinkage, cracks, and the like of the conductive member **10** due to heat. In addition, because the diamond D has a thermal conductivity higher than that of metals and has isotropy in a thermal conduction direction, disposing the diamond D in the conductive member **10** allows for improving the heat dissipation of the substrate **1**. The shape of the diamond D is, for example, a sphere or a polyhedron, and can be a polyhedron close to a sphere.

[0054] The configuration of the diamond D will be described below. Each of FIGS. **4B** to **4D** is a cross-sectional view illustrating an example of the diamond D. In the example illustrated in FIG. **4B**, the diamond D1 is a diamond particle **41**. The shape of the diamond particle **41** is, for example, a polyhedron having a cleavage plane. The diamond D can be a composite particle having a diamond particle **41** and a plating layer covering the diamond particle **41**. For example, as illustrated in FIG. **4C**, the diamond D can be a composite particle D2 having a diamond particle **41** and a nickel-plating layer **42** covering the diamond particle **41**. Meanwhile, as illustrated in FIG. **4D**, the diamond D can be a composite particle D3 having a diamond particle **41**, a nickel-plating

layer **42** covering the diamond particle **41**, and a copper plating layer **43** covering the nickel-plating layer. In particular, when the composite particle contains a diamond particle, good thermal conductivity can be obtained. In addition, with the diamond particle **41** covered with a metal film such as a plating layer, adhesion between the diamond **D** and the metal contained in the conductive member **10** is improved. In the present embodiment, the diamond **D1** composed of the diamond particle **41**, the composite particle **D2**, and the composite particle **D3** are collectively referred to as “diamond **D**”.

[0055] The particle size of each diamond **D** contained in the first region **11** is 20 μm or greater and 130 μm or less. The term “particle size” as used herein refers to the diameter of the smallest sphere that circumscribes the diamond **D**. The diamond **D** has any appropriate shape and has, for example, a hexahedron shape or an octahedron shape. If the particle size of the diamond **D** is less than 20 μm , the thermal conductivity of the diamond **D** is low, which may result in reduction in the thermal conductivity of the first region **11**. If the particle size of the diamond **D** exceeds 130 μm , the area of the diamond **D** in contact with the third region **13** is small in the vicinity of the boundary between the first region **11** and the third region **13** when the density of the diamond **D** is constant. That is, by setting the particle size of the diamond **D** to 20 μm or greater and 130 μm or less, the area of copper in the first region **11** is reduced in the vicinity of the boundary between the first region **11** and the third region **13**. Therefore, the amount of voids generated due to the dissolution of copper from the first region **11** to the third region **13** can be reduced. Thus, the particle size of the diamond **D** is preferably 20 μm or greater and 130 μm or less, more preferably greater than 20 μm and 80 μm or less.

[0056] The particle size of the diamond **D** is determined by image analysis of an observation photograph of a cross section, sorting using a sieve, or the like. For example, in scanning electron microscopy-energy dispersive X-ray spectroscopy (SEM-EDX; hereinafter occasionally referred to as “EDX”), diamond particles **41** are observed to be darker than copper. The particle size of the diamond **D** can be obtained by extracting a black region from the image and measuring the size of the region. Alternatively, the diamond **D** can be taken out by cutting out the first region **11** from the conductive member **10** and dissolving copper contained in the first region **11** with an acidic solution. The particle size of the taken-out diamond **D** is observed by, for example, observing the diamond **D** using an optical microscope, SEM (scanning electron microscopy), or SEM-EDX and measuring the size of the diamond **D**. The taken-out diamond **D** can be sieved with a mesh. The diamond **D** is sequentially sieved from a finer mesh to a coarser mesh. The particle size of the diamond **D** can be estimated from the size of the mesh when the diamond **D** passes through the mesh.

[0057] Any appropriate content of the diamond **D** can be contained in the first region **11**. Preferably, in a cross section parallel to the Z direction, the diamond **D** occupies 10% or more and 50% or less of the area of the first region **11**. When the area occupied by the diamond **D** in the cross section is less than 10%, the effect of reducing the thermal expansion coefficient of the first region **11** by the diamond **D** may be insufficient. When the area occupied by the diamond **D** exceeds 50%, the conductivity of the first region **11** may be reduced by the diamond **D**. Therefore, the area occupied by the diamond **D** in the cross section is preferably 10% or more and 50% or less, more preferably 25% or more and 45% or less, of the area of the first region **11**.

[0058] FIG. 5 is an enlarged cross-sectional view of the first region and the second region illustrated in FIG. 1.

[0059] As illustrated in FIG. 5, the first region **11** includes a first portion **11a** and a second portion **11b**. The second portion **11b** is positioned between the first portion **11a** and the second region **12** in the Z direction. When a cross section of the conductive member **10** parallel to the Z direction is observed, the area occupied by the diamond **D** per unit area in the second portion **11b** is smaller than the area occupied by the diamond **D** per unit area in the first portion **11a**.

[0060] When it is confirmed from the observation result of the cross section that the area occupied

by the diamond D per unit area in the second portion **11b** is smaller than the area occupied by the diamond D per unit area in the first portion **11a**, it is estimated that the volume occupied by the diamond D per unit volume in the second portion **11b** is smaller than the volume occupied by the diamond D per unit volume in the first portion **11a** in the first region **11** of the conductive member **10**. When the diamond D is a composite particle having a plating layer on the surface thereof, the thickness of the plating layer is very small relative to the volume of the diamond particle **41**, and thus the thickness of the plating layer is ignored.

[0061] In the first region **11**, the portion other than the diamond D is mainly made of copper. Therefore, when the area occupied by the diamond D per unit area in the second portion **11b** is smaller than the area occupied by the diamond D per unit area in the first portion **11a**, the area occupied by copper per unit area in the second portion **11b** is larger than the area occupied by copper per unit area in the first portion **11a**. In addition, when it is confirmed from the observation result of the cross section that the area occupied by copper per unit area in the second portion **11b** is larger than the area occupied by copper per unit area in the first portion **11a**, it is estimated that the volume occupied by copper per unit volume in the second portion **11b** is larger than the volume occupied by copper per unit volume in the first portion **11a** in the first region **11** of the conductive member **10**.

[0062] In the present embodiment, a difference in density of the diamond D between the first portion **11a** and the second portion **11b** is present. With such a portion with a low density of the diamond D, such as the second portion **11b**, between the first portion **11a** and the second region **12** in this way, it is possible to inhibit entry of the diamond D into the second region **12**. Inhibiting entry of the diamond D into the second region **12** allows for inhibiting the upper surface of the second region **12** from becoming uneven, and thus improving the accuracy of arrangement when arranging a light-emitting element **33** on the second region **12** as will be described below.

[0063] There can be a case in which the first region **11**, the second region **12**, and the third region **13** in the present embodiment does not have a clear boundary line that allows such regions to be visually distinguished. In this case, the first region **11**, the second region **12**, and the third region **13** are distinguished as follows.

[0064] In one example, in the cross section of the conductive member **10**, the first region **11** and the second region **12** are distinguished by the presence of the diamond D. That is, when the cross section of the conductive member **10** is observed, a region where the diamond D is present is determined as the first region **11**. A region located on the first region **11** and having no diamond D is determined as the second region **12**. In addition, in the cross section of the conductive member **10**, the first region **11** and the third region **13** are also distinguished by the presence or absence of the diamond D. That is, when the cross section of the conductive member **10** is observed, a region where the diamond D is present is determined as the first region **11**. A region located on the X-direction side with respect to the first region **11** and having no diamond D is determined as the third region **13**. Furthermore, the first region **11** and the third region **13** can be distinguished on the basis of the concentration of the active metal.

[0065] For example, SEM-EDX is used to observe the cross-section of the conductive member **10**. First, the substrate **1** is cut along the Z direction so as to include the first region **11** and the second region of the conductive member **10**. The cut surface of the substrate **1** is polished, and the polished surface is observed through SEM.

[0066] To distinguish the first region **11** and the second region **12** from each other, the diamond D is confirmed in the image obtained through SEM. In SEM, the heavier element a portion contains, the whiter the portion is observed to be. When the copper element is compared with the carbon element contained in diamond, the copper element is heavier. Therefore, in the result of observation through SEM, diamond is observed to be black as compared with copper. The first region **11** and the second region **12** are distinguished on the basis of the presence of a black region corresponding to diamond.

[0067] To distinguish the first region **11** and the third region **13** from each other, a rough boundary between the first region **11** and the third region **13** is grasped on the basis of the distribution of the diamond D in the X direction. The vicinity of the boundary is observed, and elemental mapping of the active metal is obtained through EDX. In the vicinity of the boundary between the first region **11** and the third region **13**, a change in the concentration of the active metal in the X direction is present. The first region **11** and the third region **13** are distinguished from each other on the basis of the concentration of the active metal acquired by the elemental mapping. In the first region **11**, an EDX spectrum of the active metal is not observed, and the active metal is not detected.

[0068] In the substrate **1** according to the embodiment, when the first region **11** and the third region **13** are distinguished by the above-described method, the diamond D in the first region **11** in the vicinity of the third region **13** is in contact with the third region **13**. That is, a portion of the diamond D is in contact with a region where an EDX spectrum of the active metal is observed.

[0069] In addition, the first portion **11a** and the second portion **11b** are distinguished on the basis of the area occupied by diamond or copper in the first region **11**. As described above, when the first region **11** is observed through SEM, copper and diamond have different colors. By image analysis of the observation photograph, the area of the white portion is measured as the area of copper, and the area of the black portion is measured as the area of diamond. In the first region **11**, a plurality of sections are set in the Z direction, and an area occupied by each of copper and diamond per unit area is calculated for each section. A section where the area occupied by diamond per unit area is relatively large is distinguished as the first portion **11a**. A section where the area occupied by diamond per unit area is relatively small is distinguished as the second portion **11b**.

[0070] FIG. **6** is an enlarged cross-sectional view of the ceramic member illustrated in FIG. **1**. For example, as illustrated in FIG. **6**, the width (the length in the X direction) of an upper portion **21** of the ceramic member **20** is smaller than the width (the length in the X direction) of a lower portion **22** of the ceramic member **20**. The third region **13** is disposed on the lower portion **22** in a portion other than the upper portion **21**. The thickness (the length in the Z direction) of the upper portion **21** is smaller than the thickness (the length in the Z direction) of the lower portion **22**. In this case, the third region **13** can be disposed on the lower portion **22** in a portion other than the upper portion **21**. The proportion of the third region **13** on the upper surface of the substrate **1** can be further increased. The third region **13** contains a metal, and thus the thermal conductivity of the third region **13** is higher than the thermal conductivity of the ceramic member **20**. Therefore, the heat dissipation of the substrate **1** can be further enhanced. However, the specific shape of the ceramic member **20** is not limited to the above.

[0071] FIG. **7** is a cross-sectional view illustrating a light-emitting device according to the embodiment. The substrate **1** according to the embodiment is suitably used for a light-emitting device. As illustrated in FIG. **7**, a light-emitting device **2** includes a substrate **1**, a metal layer **30**, a metal layer **31**, a bonding member **32**, a light-emitting element **33**, a phosphor member **34**, and a covering member **35**.

[0072] The metal layer **30** is disposed on the upper surface of the second region **12** and the upper surface of the third region **13**. The metal layer **31** is disposed on the lower surface of the third region **13**. The metal layers **30** and **31** include a thin film of nickel, palladium, gold, or the like.

[0073] The bonding member **32** is disposed on the metal layer **30**. In the example illustrated in FIG. **7**, the bonding members **32** are disposed at a plurality of locations to bond the substrate **1** and the light-emitting element **33**. The bonding member **32** is, for example, solder. Since the metal layer **30** is disposed on the upper surface of the ceramic member **20**, wettability for the bonding member **32** is improved.

[0074] In the example illustrated in FIG. **7**, one light-emitting device **2** includes one light-emitting element **33**. A light-emitting surface **36** is provided on the upper surface of the light-emitting element **33**. At least a pair of positive and negative element electrodes are provided on a surface of the light-emitting element **33** opposite the light-emitting surface **36**. In the example illustrated in

FIG. 7, the shape of the light-emitting surface **36** in a plan view is substantially rectangular. However, the shape of the light-emitting surface **36** in a plan view can be substantially circular or substantially elliptical, and can be polygonal such as substantially triangular or substantially hexagonal. The number of light-emitting elements **33** included in one light-emitting device **2** can be two or more.

[0075] In the example illustrated in FIG. 7, the light-emitting element **33** is disposed on the second region **12** and the ceramic member **20**, and is electrically connected to the second region **12**. The light-emitting element **33** is bonded to the conductive member **10** via the element electrodes, the metal layer **30**, and the bonding member **32**. The light-emitting element **33** has various semiconductors such as a group III-V compound semiconductor or a group II-VI compound semiconductor. The light-emitting element **33** can be a light-emitting diode (LED), or can be a laser diode (LD). As the semiconductor, preferably, a nitride-based semiconductor such as $\text{In.sub.XAl.sub.YGa.sub.1-X-YN}$ ($0 \leq X$, $0 \leq Y$, $X+Y \leq 1$) or the like is used, and InN , AlN , GaN , InGaN , AlGaN , InGaAlN , and the like can also be used. The light-emitting element **33** includes an n-type semiconductor layer, a p-type semiconductor layer, and a light-emitting layer disposed between the n-type semiconductor layer and the p-type semiconductor layer. The p-type semiconductor layer of the light-emitting element **33** is electrically connected to one of the two conductive members **10**, and the n-type semiconductor layer of the light-emitting element **33** is electrically connected to the other of the two conductive members **10**. A light emission peak wavelength of the light-emitting element **33** is preferably 400 nm or greater and 530 nm or less, more preferably 400 nm or greater and 490 nm or less, and even more preferably 440 nm or greater and 475 nm or less, from the viewpoints of light emission efficiency, excitation of the phosphor to be described below, a color mixing relationship with the light emission thereof, and the like.

[0076] The phosphor member **34** is a member having a substantially rectangular shape in a plan view, for example, and is provided so as to cover the upper surface of the light-emitting element **33**. The phosphor member **34** can be formed using a light-transmissive resin material, or an inorganic material such as ceramic or glass. As the resin material, a thermosetting resin, such as a silicone resin, a silicone modified resin, an epoxy resin, an epoxy modified resin, or a phenol resin, can be used. Particularly, a silicone resin or a modified resin thereof with good light resistance and heat resistance is preferably used. Herein, light transmissivity corresponds to preferably 60% or more of the light from the light-emitting element **33** being transmitted. Further, a thermoplastic resin such as a polycarbonate resin, an acrylic resin, a methyl pentene resin, or a polynorbornene resin can be used as the phosphor member **34**. Furthermore, the phosphor member **34** contains a light diffusion substance or a phosphor that converts the wavelength of at least a portion of light from the light-emitting element **33**. For example, the phosphor member **34** can be a resin material, ceramic, glass, or the like containing a phosphor, a sintered body of a phosphor, or the like. Further, the phosphor member **34** can be a multi-layer member in which a resin layer containing a phosphor and a light diffusion substance is disposed on the upper surface or the lower surface of a molded body formed of resin, ceramic, glass, or the like.

[0077] As the phosphor, an yttrium aluminum garnet-based phosphor (for example, $(\text{Y,Gd}).\text{sub.3}(\text{Al,Ga}).\text{sub.5O}.\text{sub.12}:\text{Ce}$), a lutetium aluminum garnet-based phosphor (for example, $\text{Lu}.\text{sub.3}(\text{Al,Ga}).\text{sub.5O}.\text{sub.12}:\text{Ce}$), a terbium aluminum garnet-based phosphor (for example, $\text{Tb}.\text{sub.3}(\text{Al,Ga}).\text{sub.5O}.\text{sub.12}:\text{Ce}$), a CCA-based phosphor (for example, $\text{Ca}.\text{sub.10}(\text{PO}.\text{sub.4}).\text{sub.6Cl}.\text{sub.2}:\text{Eu}$), an SAE-based phosphor (for example, $\text{Sr}.\text{sub.4Al}.\text{sub.14O}.\text{sub.25}:\text{Eu}$), a chlorosilicate-based phosphor (for example, $\text{Ca}.\text{sub.8MgSi}.\text{sub.4O}.\text{sub.16Cl}.\text{sub.2}:\text{Eu}$), a silicate-based phosphor (for example, $(\text{Ba,Sr,Ca,Mg}).\text{sub.2SiO}.\text{sub.4}:\text{Eu}$), an oxynitride-based phosphor β - SiAlON -based phosphor (for example, $(\text{Si,Al}).\text{sub.3}(\text{O,N}).\text{sub.4}:\text{Eu}$) or an α - SiAlON -based phosphor (for example, $\text{Ca}(\text{Si,Al}).\text{sub.12}(\text{O,N}).\text{sub.16}:\text{Eu}$), a nitride-based phosphor such as an LSN-based phosphor (for example, $(\text{La,Y}).\text{sub.3Si}.\text{sub.6N}.\text{sub.11}:\text{Ce}$), a BSESN-based phosphor (for example,

(Ba,Sr).sub.2Si.sub.5N.sub.8:Eu), an SLA-based phosphor (for example, SrLiAl.sub.3N.sub.4:Eu), a CASN-based phosphor (for example, CaAlSiN.sub.3:Eu), or an SCASN-based phosphor (for example, (Sr,Ca)AlSiN.sub.3:Eu), a fluoride-based phosphor such as a KSF-based phosphor (for example, K.sub.2SiF.sub.6:Mn), a KSAF-based phosphor (for example, K.sub.2(Si.sub.1-xAl.sub.x)F.sub.6-x:Mn, where x satisfies $0 < x < 1$), or an MGF-based phosphor (for example, 3.5MgO.Math.0.5MgF.sub.2.Math.GeO.sub.2:Mn), a quantum dot having a perovskite structure (for example, (Cs,FA,MA)(Pb,Sn)(F,Cl,Br,I).sub.3, where FA and MA represent formamidinium and methylammonium, respectively), a group II-VI quantum dot (for example, CdSe), a group III-V quantum dot (for example, InP), a quantum dot having a chalcopyrite structure (for example, (Ag,Cu)(In,Ga)(S,Se).sub.2), or the like can be used.

[0078] In the example illustrated in FIG. 7, the light-emitting element **33** emits blue light, and the phosphor member **34** includes a phosphor that absorbs blue light and emits red light and a phosphor that absorbs blue light and emits green light. Alternatively, the phosphor member **34** can include a phosphor that absorbs blue light and emits yellow light.

[0079] The covering member **35** is disposed so as to cover the lateral surface of the light-emitting element **33** and the lateral surface of the phosphor member **34**. The covering member **35** covers the lateral surfaces of the light-emitting element **33** and the phosphor member **34** directly or indirectly. The upper surface of the phosphor member **34** is exposed from the covering member **35** and corresponds to the light-emitting surface of the light-emitting device **2**. The covering member **35** is further disposed between the metal layer **30** and the light-emitting element **33** and between the ceramic member **20** and the light-emitting element **33**. The covering member **35** reflects light emitted toward the covering member **35** from the light-emitting element **33**. The covering member **35** is integrally formed of, for example, a white resin material. The covering member **35** is preferably constituted by a member having a high light reflectivity in order to improve light extraction efficiency. As the covering member **35**, an organic material such as a resin containing a light-reflective material such as white pigment, for example, can be used. Alternatively, the covering member **35** can be a light-reflective member composed of an inorganic material containing boron nitride and alkali metal silicate, for example. In this case, titanium oxide or zirconium oxide can be further contained.

[0080] Examples of the light reflective material include titanium oxide, zinc oxide, magnesium oxide, magnesium carbonate, magnesium hydroxide, calcium carbonate, calcium hydroxide, calcium silicate, magnesium silicate, barium titanate, barium sulfate, aluminum hydroxide, aluminum oxide, zirconium oxide, and silicon oxide. One type of these can be used alone, or two or more types of these can be used in combination. The resin material as the organic material is preferably a material in which a resin material including, as a main component, a thermosetting resin, such as an epoxy resin, an epoxy modified resin, a silicone resin, a silicone modified resin, a phenol resin, or the like is used as a base material. The covering member **35** can be constituted by a member having light transmissivity or absorbency for visible light as necessary.

[0081] When the light-emitting device **2** includes a plurality of light-emitting elements **33** and phosphor members **34** disposed on the light-emitting elements **33**, the covering member **35** can be disposed between the adjacent light-emitting elements **33** and between the adjacent phosphor members **34** to integrally cover the plurality of light-emitting elements **33** and the plurality of phosphor members **34**.

[0082] Advantages that can be provided by the embodiment of the present disclosure will be described below.

[0083] When light is emitted from the light-emitting device **2**, heat is generated in the light-emitting element **33**. The heat of the light-emitting element **33** is transmitted to the conductive member **10**, the ceramic member **20**, and the like. The temperatures of the conductive member **10**, the ceramic member **20**, and the light-emitting element **33** change depending on whether or not heat is generated by the light-emitting element **33**. When the temperature changes, thermal

expansion or thermal contraction occurs in each of the conductive member **10**, the ceramic member **20**, and the light-emitting element **33**.

[0084] In the present embodiment, the first region **11** is provided in the conductive member **10**. The first region **11** contains copper and diamond. The thermal expansion coefficient of the first region **11** containing copper and diamond is smaller than the thermal expansion coefficient of the second region **12** not containing diamond. With the conductive member **10** including the first region **11**, the thermal expansion amount and the thermal contraction amount of the conductive member **10** can be reduced. Accordingly, stress generated between the conductive member **10** and the ceramic member **20** and stress generated between the conductive member **10** and the light-emitting element **33** during thermal expansion or thermal contraction can be reduced. As a result, the occurrence of breakage of the light-emitting element **33**, malfunction of the light-emitting device **2**, and the like due to temperature rise can be reduced.

[0085] According to the present embodiment, the substrate **1** including the conductive member **10** in which a difference in thermal expansion coefficient from the ceramic member **20** or the light-emitting element **33** can be provided. In addition, by using the substrate **1** according to the embodiment in the light-emitting device **2**, the light-emitting device **2** having high reliability with respect to temperature changes can be provided.

[0086] FIGS. **8A** to **11C** are each a schematic view illustrating a step of manufacturing the substrate according to the embodiment.

[0087] The steps of manufacturing the substrate according to the embodiment includes a step of providing a sintered body having a plurality of protrusions, a step of disposing a ceramic member, and a step of bonding the ceramic member and the sintered body.

[0088] First, the step of providing a sintered body having a plurality of protrusions will be described. In the step of providing a sintered body having a plurality of protrusions, a first raw material **55** containing copper powder and diamond is disposed inside recesses **52** of a jig **50** having an upper surface **51** and the recesses **52** recessed from the upper surface **51**. A second raw material **57** containing copper powder is disposed on the first raw material **55** and the upper surface **51** of the jig **50**, and the first raw material **55** and the second raw material **57** are sintered.

[0089] The jig **50** illustrated in FIG. **8A** is used to manufacture the substrate **1**. FIG. **8B** is an enlarged cross-sectional view of a part of the jig **50**. As illustrated in FIG. **8B**, the jig **50** is a member having the upper surface **51** and the recesses **52** recessed from the upper surface **51**.

[0090] As illustrated in FIG. **8C**, a mold release material **53** is applied by spraying to the upper surface **51** and the inner surface of each of the recesses **52** of the prepared jig **50**. The mold release material **53** contains carbon powder or boron nitride powder. As illustrated in FIG. **8D**, the first raw material **55** is disposed inside each of the recesses **52**. The first raw material **55** contains, for example, 60 wt % or more and 85 wt % or less of copper powder and 15 wt % or more and 40 wt % or less of the diamond **D**. The particle size of the diamond **D** is 20 μm or greater and 130 μm or less. It is preferable that the first raw material **55** is disposed such that almost no gap inside the recess **52** is present. Then, the upper surface of the mold release material **53** disposed on the upper surface **51** of the jig **50** is made flush with the upper surface of the first raw material **55** disposed inside the recess **52**.

[0091] As illustrated in FIG. **9A**, the jig **50** is disposed inside a tubular body **56**, and the second raw material **57** is disposed on the upper surface of the mold release material **53** and the upper surface of the first raw material **55**. The second raw material **57** contains copper powder. In the present embodiment, the second raw material **57** does not contain diamond. The second raw material **57** is disposed on the upper surface **51** and on the first raw material **55**. FIG. **9B** is an enlarged cross-sectional view of a part of FIG. **9A**. As illustrated in FIG. **9B**, a weight **58** is placed on the second raw material **57**, and the first raw material **55** and the second raw material **57** are pressurized at a 10 MPa or more and a 50 MPa or less. At this time, a part of the second raw material **57** enters the inside of each of the recesses **52**. Thus, the first raw material **55** and the second raw material **57** are

disposed inside the recess **52**, and the thickness of the first raw material **55** disposed inside the recess **52** is greater than the thickness of the second raw material **57** disposed inside the recess **52**. That is, the proportion of the first raw material **55** is higher than the proportion of the second raw material **57** inside the recess **52**.

[0092] The mean particle diameter of the copper powder of the first raw material **55** and the mean particle diameter of the copper powder of the second raw material **57** are preferably 1 μm or greater and 8 μm or less. When the mean particle diameter of the copper powder is less than 1 μm , the proportion of the surface area of the copper powder increases. The surface of the copper powder is easily oxidized. Therefore, in a case in which the surface of the copper powder is oxidized, the area of the copper oxide also increases, and thus there is a possibility that the sinterability decreases. When the mean particle diameter of the copper powder exceeds 8 μm , the copper powder is too large to densely cover the surface of the diamond **D**, and the density and thermal conductivity of the sintered body may be reduced. Thus, the mean particle diameter of the copper powder is preferably 1 μm or greater and 8 μm or less.

[0093] In this state, the first raw material **55** and the second raw material **57** are sintered while applying a pressure of 30 MPa or more and 90 MPa or less. For example, the sintering temperature is set to 600° C. or more and 750° C. or less. The copper powder contained in the first raw material **55** and the second raw material **57** is sintered to obtain a sintered body **60** illustrated in FIG. **9C**. The sintered body **60** is removed from the jig **50**. The sintered body **60** includes a base portion **61** having a flat plate shape and a plurality of protrusions **62** disposed on the base portion **61**. The size and shape of the protrusion **62** correspond to the size and shape of the recess **52**. In the present embodiment, the protrusions **62** each have a rectangular shape in a plan view and a three-dimensional shape of a rectangular parallelepiped.

[0094] As illustrated in FIG. **9B**, the first raw material **55** and the second raw material **57** are preferably sintered while being pressurized. This makes it possible to improve the density and thermal conductivity of the sintered body **60** to be fabricated.

[0095] In the sintered body **60** of the present embodiment, the base portion **61** contains copper as a main component. A region **62a** of the protrusion **62** contains copper and diamond. A region **62b** of the protrusion **62** contains copper as a main component. The region **62b** is located between the base portion **61** and the region **62a**. The region **62a** is formed by sintering the first raw material **55**. The base portion **61** and the region **62b** are formed by sintering the second raw material **57**. Therefore, in the sintered body **60** of the present embodiment, the base portion **61** and the region **62b** do not contain diamond.

[0096] The difference between the area occupied by diamond per unit area of the first portion **11a** and that of the second portion **11b** of the substrate **1** results from the fact that the second raw material **57** is pressed toward the first raw material **55** as illustrated in FIG. **9B**. When the pressure is applied to the second raw material **57**, a part of the second raw material **57** enters the recess **52** of the jig **50**. A part of the second raw material **57** is mixed with a part of the first raw material **55**. The first raw material **55** contains diamond, but the second raw material **57** does not contain diamond. Therefore, the density of diamond is lower in the portion where the first raw material **55** and the second raw material **57** are mixed than that in the portion where such raw materials are not mixed. By sintering the first raw material **55** and the second raw material **57** in this state, the first portion **11a** and the second portion **11b** of the substrate **1** are formed.

[0097] Subsequently, the step of disposing a ceramic member is performed. In the step of disposing a ceramic member, a ceramic member is disposed on the lateral surfaces of each of the plurality of protrusions **62** of the sintered body **60** removed from the jig **50** via an active metal brazing material.

[0098] As illustrated in FIG. **10A**, a ceramic member **70** having holes **71** is prepared. In a plan view, the outer shape of the protrusion **62** of the sintered body **60** is similar to the outer shape of the hole **71**. The ceramic member **70** has a plurality of holes **71** penetrating from the upper surface to

the lower surface. The position and size of the hole **71** corresponds to the position and size of the protrusion **62**. In the present embodiment, a thickness **t1** (the size in the Z direction) of the ceramic member **70** is greater than a thickness **t2** of the protrusion **62**.

[0099] As illustrated in FIG. **10B**, an active metal brazing material **63** is adhered to the periphery of the protrusions **62** by printing so as to be flush with the upper surfaces of the protrusions **62**. The active metal brazing material only needs to be disposed around the protrusions **62**, and can be adhered by syringe filling. At this time, the active metal brazing material **63** can be adhered to the upper surfaces of the protrusions **62**. The active metal brazing material **63** contains copper and an active metal element. The active metal element is at least one selected from the group consisting of titanium, hafnium, zirconium, niobium, cerium, and magnesium. The active metal brazing material **63** can further contain silver, tin, indium, or the like. In particular, since the melting point of the active metal brazing material **63** is lowered when the active metal brazing material **63** contains silver, the sintering temperature can be lowered in the step of bonding the ceramic member and the sintered body to be described below.

[0100] As illustrated in FIG. **10C**, the protrusions **62** of the sintered body **60** having the active metal brazing material **63** adhered to the lateral surface of each thereof are inserted into the corresponding holes **71** of the ceramic member **70**. In the present embodiment, the outer peripheral surface of the protrusion **62** and the inner peripheral surface of the hole **71** are spaced apart from each other to such an extent that the active metal brazing material **63** can be disposed between the protrusion **62** and the hole **71**. Thus, the ceramic member **70** is disposed on the lateral surface of each of the plurality of protrusions **62** via the active metal brazing material **63**. The active metal brazing material **63** is also disposed between the upper surface of the base portion **61** and the lower surface of the ceramic member **70**.

[0101] Subsequently, the step of bonding the ceramic member **70** and the sintered body **60** is performed by sintering the active metal brazing material **63**.

[0102] In the step of bonding the ceramic member and the sintered body, as illustrated in FIG. **11A**, an active metal brazing material **63** is further adhered onto the protrusions **62** and the ceramic member **70** by printing, and a copper plate **80** is disposed thereon. The size of the copper plate **80** in a plan view is substantially the same as the size of the base portion **61** in a plan view. The copper plate **80** preferably has a thickness equivalent to that of the base portion **61** of the sintered body **60** from the viewpoint of reducing warpage during sintering. The thickness of the copper plate **80** is, for example, 50 μm or greater and 300 μm or less. A layered body composed of the sintered body **60**, the active metal brazing material **63**, the ceramic member **70**, and the copper plate **80** is sintered at a temperature of 700° C. or greater and to 1200° C. or less. At this time, if the diamond **D** is a composite particle having the diamond particle **41** and a plating layer covering the diamond particle **41**, the plating layer can be melted. The melted plating layer becomes a part of the region **62a** (the conductive member **10** to be described below) of the protrusion **62**.

[0103] By the sintering, the active metal brazing material **63** is melted and solidified to form a bonding layer **65** as illustrated in FIG. **11B**. The sintered body **60**, the ceramic member **70**, and the copper plate **80** are bonded to each other by the bonding layer **65** to fabricate a bonded body **90**. The step of bonding the ceramic member and the sintered body further includes a step of removing a portion of the sintered body **60** other than the plurality of protrusions **62** after sintering the active metal brazing material **63**. As illustrated in FIG. **11C**, both surfaces of the bonded body **90** are ground. By grinding both surfaces, the base portion **61** of the sintered body **60**, a part of the bonding layer **65**, and the copper plate **80** are removed. Of the sintered body **60**, only the plurality of protrusions **62** remain. In FIG. **11C**, the bonded body **90** is illustrated upside down with respect to FIG. **11B**.

[0104] In the present embodiment, a part of the sintered body **60** is ground and removed as in the steps illustrated in FIGS. **11B** and **11C**. At this time, the base portion **61** and a part of the region **62b** of the protrusion **62** are ground. In the present embodiment, as illustrated in FIG. **10A**, the

thickness **t1** of the ceramic member **70** fitted to the sintered body **60** is greater than the thickness **t2** of the protrusion **62**. Therefore, when the copper plate **80** is ground in the subsequent step illustrated in FIG. **11C**, the upper surface and the lower surface of the ceramic member **70** serve as stoppers. At this time, the protrusions **62** located on the inner side between the upper surface and the lower surface of the ceramic member **70** in the Z direction are hardly ground. The protrusion **62** at the region **62b** has been formed by sintering the second raw material **57** disposed on the first raw material **55** while being pressurized. Since the protrusion **62** does not contain diamond, the protrusion **62** is easily ground. Diamond is harder than copper. Therefore, if a part of the sintered body **60** made of the first raw material **55** is ground, a difference in grinding amount occurs between diamond and copper. As a result, the unevenness of the ground surface increases, reducing flatness. On the other hand, the second raw material **57** does not contain diamond, and therefore can be ground more uniformly. Since diamond is not contained in the region **62b** of the protrusion **62**, the unevenness of the ground surface of the region **62b** can be reduced, improving flatness.

[0105] On the opposite side of the base portion **61**, a part of the bonding layer **65** adjacent to the protrusion **62** remaining without being ground is a portion to be the third region **13** of the substrate **1**. Since the bonding layer **65** does not contain diamond, the bonding layer **65** is easily ground.

[0106] The bonded body **90** including the substrate according to the embodiment is fabricated through the steps up to FIG. **11C**. The obtained bonded body **90** can be used as the substrate **1**, or a plurality of substrates **1** can be manufactured by dividing the bonded body **90** into individual pieces so as to include at least two sintered bodies **60** (protrusions **62**). In the bonded body **90**, the sintered body **60** and the bonding layer **65** correspond to the conductive member **10** illustrated in FIG. **1**. The region **62a** corresponds to the first region **11**. The region **62b** corresponds to the second region **12**. The bonding layer **65** corresponds to the third region **13**. The ceramic member **70** corresponds to the ceramic member **20**.

[0107] FIGS. **12A** to **12C** are each a schematic view illustrating a method of manufacturing the light-emitting device according to the embodiment.

[0108] The method of manufacturing the light-emitting device **2** includes a step of manufacturing a substrate by the above-described manufacturing method, and a step of disposing one light-emitting element on each of two sintered bodies (corresponding to the protrusions **62** before being removed) among a plurality of sintered bodies (corresponding to the protrusions **62** before being removed) of the substrate.

[0109] Hereinafter, a method of manufacturing the light-emitting device **2** in which the bonded body **90** before being divided into individual pieces is used as the substrate will be described. In the present embodiment, the method further includes a step of forming a metal layer on the plurality of protrusions and the ceramic member after the step of disposing a ceramic member. First, electroless plating (displacement plating) is performed on the bonded body **90**. As a result, as illustrated in FIG. **12A**, a metal layer **30** is formed on the upper surface of the sintered body **60** and the upper surface of the bonding layer **65**, and a metal layer **31** is formed on the lower surface of the bonding layer **65**. In the present embodiment, since the surface of the sintered body **60** exposed on the upper surface of the bonded body **90** and the lower surface of the bonding layer **65** exposed on the lower surface of the bonded body **90** become the patterns of the metal layer **30** and the metal layer **31**, respectively, a mask is not required during electroless plating. Electrolytic plating can be performed with the bonded body **90** provided with wiring lines for supplying power to the surface of the sintered body **60** exposed on the upper surface of the bonded body **90** and the lower surface of the bonding layer **65** exposed on the lower surface of the bonded body **90**.

[0110] As illustrated in FIG. **12B**, light-emitting elements **33** are bonded onto the metal layer **30** via bonding members **32**. At this time, one light-emitting element **33** is disposed on two sintered bodies **60** among the plurality of sintered bodies **60** of the bonded body **90** (substrate) such that the positive and negative element electrodes of the light-emitting element **33** are connected to the two sintered bodies **60**, respectively. A phosphor member **34** is disposed on each light-emitting element

33. The light-emitting element **33** and the phosphor member **34** can be directly bonded to each other, or can be indirectly bonded to each other with an adhesive member. A white resin is disposed in the gap between the metal layers **30** and the gap between the bonded body **90** and the metal layer **30**, and then cured. As a method of molding a white resin, for example, compression molding or transfer molding can be used. When a white resin is provided on the upper surface of the phosphor member **34**, the white resin is removed by grinding to expose the phosphor member **34**. Thus, as illustrated in FIG. **12C**, a covering member **35** covering the lateral surfaces of the light-emitting element **33** and the phosphor member **34** is formed. Alternatively, the covering member **35** can be composed of, for example, a mixture containing boron nitride and alkali metal silicate, instead of the white resin. The mixture can be fabricated by mixing mixed powder of boron nitride powder and silicon oxide powder with an alkaline solution (e.g., potassium hydroxide), followed by heat curing. In the case of potassium hydroxide as the alkaline solution, heat curing causes the silicon oxide and potassium hydroxide to react with each other to form potassium silicate, which is an alkali metal silicate. Boron nitride is a member that can reduce shrinkage of the mixture during heat curing. Aluminum oxide can be used instead of boron nitride.

[0111] The present embodiment further includes a step of dividing the substrate **1** into a plurality of pieces such that the light-emitting device **2** includes at least one light-emitting element **33**.

Specifically, as illustrated in FIG. **12D**, the covering member **35** and the bonded body **90** are cut along broken lines L located between the adjacent light-emitting elements **33**. The light-emitting device **2** according to the present embodiment is manufactured through the above steps.

Variations of Substrate **1**

[0112] FIGS. **13A** to **13C** are each a cross-sectional view illustrating a substrate according to a variation of the embodiment. The specific configuration of the substrate according to the embodiment is not limited to the example illustrated in FIGS. **1** to **3**. For example, in a substrate **1A** illustrated in FIG. **13A**, the length of the ceramic member **20** in the X direction is constant over the Z direction. Since it is not necessary to form the ceramic member **20** into a protruding shape in the substrate **1A**, the ceramic member **20** can be easily processed, improving productivity. A substrate **1B** illustrated in FIG. **13B** further includes a ceramic member **20a** and a ceramic member **20b**, as compared with the substrate **1**. The ceramic member **20a** and the ceramic member **20b** are disposed at both ends of the substrate **1B** in the X direction, respectively. The conductive members **10** and the ceramic member **20** are located between the ceramic member **20a** and the ceramic member **20b** in the X direction. In the substrate **1B**, there is no diamond at the cut position when the bonded body **90** is divided into individual pieces, so that the bonded body **90** is easily divided into individual pieces. A substrate **1C** illustrated in FIG. **13C** differs from the substrate **1B** in that the third region **13** is disposed below each of the ceramic members **20a** and **20b**. In the substrate **1C**, there is no diamond at the cut position, as in the substrate **1B**, so that the bonded body **90** is easily divided into individual pieces. In addition, with the third region **13** of the conductive member **10** is exposed on the lower surface side of the lateral surface of the substrate **1C**, solder creeps up the lateral surface of the substrate **1C** to form a fillet when the substrate **1C** is mounted on the mounting substrate with solder. Therefore, the heat dissipation of the substrate **1C** can be improved.

[0113] As illustrated in FIGS. **13A** to **13C**, the specific configuration of the substrate according to the embodiment can be appropriately changed as long as the first region **11** is provided in the conductive member **10**.

[0114] The substrate, the light-emitting device, the method of manufacturing a substrate, and the method of manufacturing a light-emitting device of the present disclosure can provide a substrate that can further reduce a difference in thermal expansion coefficient between a conductive member and a ceramic member, and a light-emitting device including the substrate. Therefore, the present disclosure is suitably applicable to a substrate for use in light sources for in-vehicle use, light sources for illumination use, light sources for various indicators, light sources for displays, light sources for liquid crystal display backlights, and light-emitting devices for traffic signals, in-

vehicle components, channel letters for signboards, and the like. However, the substrate, the light-emitting device, the method of manufacturing a substrate, and the method of manufacturing a light-emitting device of the present disclosure can be applied to a substrate including a conductive member and a ceramic member for use in various applications.

[0115] The above-described embodiment has been presented by way of example embodying the present disclosure, and is not intended to limit the scope of the present disclosure. For example, additions, deletions, or changes of some components or steps in the aforementioned embodiment are also included in the present disclosure. The aforementioned embodiments can be implemented in combination with each other.

Claims

1. A substrate comprising: a conductive member comprising a first region containing copper and diamond, a second region located on the first region and contains copper as a main component, a first direction being a direction from the first region toward the second region, and a third region; and a ceramic member disposed away from the first region and the second region in a second direction perpendicular to the first direction, wherein the third region is located between the first region and the ceramic member and between the second region and the ceramic member, the third region contains copper and at least one selected from the group consisting of titanium, hafnium, zirconium, niobium, cerium, and magnesium, the diamond disposed in the first region in a vicinity of the third region is in contact with the third region, and in a cross section parallel to the first direction, the first region comprises a first portion and a second portion, the second portion is located between the first portion and the second region in the first direction, and an area occupied by the diamond per unit area in the second portion is smaller than an area occupied by the diamond per unit area in the first portion.
2. The substrate according to claim 1, wherein the third region includes a portion located below the first region.
3. The substrate according to claim 1, wherein a particle size of the diamond is 20 μm or greater and 130 μm or less.
4. The substrate according to claim 1, wherein a particle size of the diamond is greater than 20 μm and less than 80 μm .
5. The substrate according to claim 1, wherein in the cross section, the diamond occupies 10% or more and 50% or less of an area of the first region.
6. The substrate according to claim 1, wherein a thickness of the first region in the first direction is greater than a thickness of the second region in the first direction.
7. The substrate according to claim 1, wherein a length of the first region in the second direction and a length of the second region in the second direction are each greater than a length of the ceramic member in the second direction.
8. The substrate according to claim 1, wherein a length of an upper portion of the ceramic member in the second direction is less than a length of a lower portion of the ceramic member in the second direction.
9. The substrate according to claim 1, wherein the third region further contains silver.
10. A light-emitting device comprising: the substrate according to claim 1; and a light-emitting element provided on the second region and the ceramic member, the light-emitting element being electrically connected to the second region.
11. A method of manufacturing a substrate, the method comprising: providing a sintered body having a plurality of protrusions by: disposing a first raw material inside recesses of a jig, the first raw material containing copper powder and diamond, the jig having an upper surface and the recesses each being recessed from the upper surface, disposing a second raw material containing copper powder on the first raw material and the upper surface of the jig, and sintering the first raw

material and the second raw material; disposing, via an active metal brazing material, a ceramic member on at least one lateral surface of each of the plurality of protrusions of the sintered body removed from the jig; and bonding the ceramic member and the sintered body by sintering the active metal brazing material.

12. The method of manufacturing a substrate according to claim 11, further comprising forming a metal layer on the plurality of protrusions and the ceramic member after the disposing of the ceramic member.

13. The method of manufacturing a substrate according to claim 11, wherein the bonding of the ceramic member and the sintered body further comprises removing a portion of the sintered body other than the plurality of protrusions after the sintering of the active metal brazing material.

14. The method of manufacturing a substrate according to claim 11, wherein a mean particle diameter of the copper powder of the first raw material and a mean particle diameter of the copper powder of the second raw material are 1 μm or greater and 8 μm or less.

15. The method of manufacturing a substrate according to claim 11, wherein a particle size of the diamond is 20 μm or greater and 130 μm or less.

16. The method of manufacturing a substrate according to claim 11, wherein in the providing of the sintered body, the first raw material and the second raw material are sintered while applying a pressure.

17. The method of manufacturing a substrate according to claim 11, wherein a thickness of the first raw material disposed inside each of the recesses is greater than a thickness of the second raw material disposed inside corresponding one of the recesses.

18. The method of manufacturing a substrate according to claim 11, wherein a thickness of the ceramic member is greater than a thickness of each of the plurality of protrusions.

19. A method of manufacturing a light-emitting device, the method comprising: manufacturing a substrate by the method of manufacturing a substrate according to claim 11; and disposing one light-emitting element on each of two protrusions among the plurality of protrusions of the sintered body of the substrate.

20. The method of manufacturing a light-emitting device according to claim 19, further comprising dividing the substrate into a plurality of pieces such that at least one light-emitting element is comprised in each of the plurality of pieces.
